Boundary and initial conditions in EXHALE (EXHALE)

This memo documents the boundary conditions and the initial conditions of the EXHALE / EXHALE hydrodynamics solver at the level of the actual implementation: what is imposed, where in the code, why it has the form it does, and what dynamical behavior it produces (in particular the "breathing base"). File references are relative to the repository root; the structure is inherited from ATES v2 (Caldiroli et al.) unless noted as an EXHALE extension.

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1. Overview: where and when the BCs act

1.1 Ghost-cell layout

The radial grid carries N computational cells ($j = 1,\dots,N$) plus $N_g = 2$ ghost cells on each side, so all state arrays are dimensioned (1-Ng : N+Ng), i.e. $j = -1, 0$ below the base and $j = N{+}1, N{+}2$ above the top (src/modules/init/parameters.f90, define_grid.f90).

2026-08-15: the cell count is a runtime quantity. N used to be a compile-time parameter fixed at 500. It is now the module variable integer :: N = 500 of parameters.f90, set by the optional input.inp key Grid cells: <N> (input_read.f90); an input.inp without the key keeps 500 and reproduces its earlier result. The grid-sized module arrays became allocatable (lower bound 1-Ng unchanged) and are allocated by allocate_grid_arrays once parsing has resolved the key, so nothing below is affected other than that N may now differ from 500. Values under 10 are refused. Two consequences appear later: the base refinement pairs of §2.10, and the restart record count of §5.5.

The first interior cell $j = 1$ sits at the lower boundary radius $r = 1$ (lengths are normalized to the planetary radius R_0); the last interior cell $j = N$ sits at $r_{\max}$, which is either the Roche/Hill truncation $r_{\max} = (3M_*/M_p)^{-1/3}\,\tilde{a}$ in Roche mode or a user-chosen outer radius in spherical mode (input_read.f90).

All boundary conditions are applied to the *ghost cells only*; the interior cells $1,\dots,N$ are never overwritten. The fluxes at the two boundary interfaces then emerge from the same HLLC Riemann solver used everywhere else, with the ghost states as the outside input. In this sense every BC in the code is a *weak* (flux-mediated) condition: even the "Dirichlet" base values act on the interior only through the Riemann problem at the interface $j = 1/2$.

1.2 The three entry points (Apply_BC.f90)

All BC logic lives in one module, src/modules/states/Apply_BC.f90:

- Apply_BC(u): conservative-variable wrapper. It converts the *interior* cells $U \to W$, calls Apply_BC_W, and converts the four ghost cells back. Called after every Runge–Kutta stage in the main loop (EXHALE_main.f90), before every residual evaluation, and by the steady-state Newton solver each time it unpacks a trial state. It *writes the ghosts and nothing else*: the interior it is handed comes back bit for bit. The earlier version round-tripped the interior as well, and that round trip is not the identity in floating point (it recomputes the kinetic

energy as $\frac{1}{2}\rho v^2$ where the input held $\frac{1}{2}(\rho v)^2/\rho$), so `Apply_BC` was not idempotent and a second call moved every interior cell at the last bit.

- `Apply_BC_W(W)`: the assignment of the four ghost states. The lower pair comes from `base_boundary_states` (§2.), which also stores the base face state and the state at the next face down for `Rec_BC`; the upper pair comes from `free_outflow_ghost` (§3.).
- `Rec_BC(WL,WR,...)`: the boundary states of the *reconstructed* left/right interface states after PLM or WENO3 reconstruction (`Reconstruction.f90`). At the base it does not re-derive anything: it reads the face state `Apply_BC_W` stored, so the ghost path and the face path are one answer from one function. At the top it extrapolates as the scheme requires.

In the Newton steady-state solver the ghost cells are deliberately *not* unknowns: the residual is defined on the interior state only, and `Apply_BC` regenerates the ghosts from the interior at every function evaluation (`src/modules/time_step/steady_newton.f90`). The BCs are therefore identical between the time-marching and Newton phases by construction.

1.3 Normalization constants tied to the base

The lower BC is intertwined with the code-wide normalization (`input_read.f90`):

$$\mu = \frac{1+4Y}{1+Y}\, m_{\rm H}, \qquad v_0 = \sqrt{\frac{k_B T_0}{m_{\rm H}}}, \qquad p_0 = n_0 m_{\rm H} v_0^2 = n_0 k_B T_0, \qquad b_0 = \frac{G M_p m_{\rm H}}{k_B T_0 R_0}, \tag{1}$$

where $Y \equiv n_{\rm He}/n_{\rm H}$ is the `He/H number ratio` from `input.inp`, n_0 is the user-specified total *nuclei* number density at the base, and T_0 the base temperature. (The code variable `mu` is $m_{\rm H}$ itself; the composition factor appears separately as `rho_bc`, see below.) Densities are stored as mass densities in units of $n_0 m_{\rm H}$, number densities in units of n_0, pressures in units of p_0, temperatures in units of T_0, velocities in units of the isothermal hydrogen sound speed v_0.

The equation-of-state closure used to convert between the hydrodynamic pressure and the temperature handed to the ionization/radiation module is the dimensionless ideal-gas law including free electrons (`composition.f90`):

$$T = \frac{p}{n_{\rm tot} + n_e}, \tag{2}$$

with $n_{\rm tot}$ the total *atomic* (nuclei) number density and n_e from `calc_ne`. Under the default `eos_metals 1` policy (2026-06-13) both include the trace metals (their nuclei in $n_{\rm tot}$ and their electrons in n_e) consistently with the charge balance inside the ionization solver; `eos_metals 0` in `metals.inp` restores the legacy H/He-only budget (§2.12).

2. Lower boundary: a characteristic condition at the base face

The lower boundary is imposed *at the face* $r_{\rm edg}(0)$, and the number of conditions it states is decided by the sign and the magnitude of the Mach number there. One module carries all of it, `src/modules/states/base_boundary.f90`, and one entry point, `base_boundary_states`, returns the face state, the two ghost cell averages and the point state at the next face down from the first interior *cell average*, so the ghost path of `Apply_BC_W` and the face path of `Rec_BC` are two readers of one answer and cannot drift apart.

This replaces a component-wise ghost-cell closure, in which a Dirichlet density, a one-way velocity valve and an isothermal ghost pressure were written into a ghost *cell* and then reused by `Rec_BC` as the state *at the face*; the two locations differ by half a cell of hydrostatic stratification, $\Delta r/(2H)$, and dividing that difference by Δr to make a momentum flux left an imbalance that was a fixed fraction of the local weight ρg at every resolution. READ from the module header, which records the measurement: 0.2514, 0.2507, 0.2503 and 0.2502 of ρg at cell 1 over an eightfold refinement of the hot-Uranus base grid, against an interior truncation error of 4.9×10^{-6}, driving a first-order velocity artifact of −247.1, −123.2, −61.3 and −30.5 cm s^{-1} over the same ladder.

2.1 How many conditions, and which

At $r_{\rm edg}(0)$ the eigenvalues of the Euler system are $v-c$, v and $v+c$, and a wave enters the domain when its eigenvalue is positive: the radial coordinate increases outward, so gas entering at the base moves outward and inflow is $v > 0$. The count follows:

Face state	Branch	Conditions
$0 < v < c$	subsonic inflow	2 reservoir, 1 interior
$-c < v < 0$	flow reversal	1 reservoir, 2 interior
$v > c$	supersonic inflow	3 reservoir, 0 interior
$v < -c$	supersonic outflow	0 reservoir, 3 interior

The interior relation is the linearized $v-c$ compatibility condition in its LODI form (Thompson 1987; Poinsot & Lele 1992; as stated in Carlson 2011, NASA/TM-2011-217181, §2),

$$p_b - \rho_i c_i v_b \;=\; p_i - \rho_i c_i v_i, \tag{3}$$

with the subscript i denoting the first interior state continued to the face. The closed-form invariant $v - 2c/(\gamma-1)$ is deliberately *not* used: it assumes one constant γ, while the caloric EOS gives every cell its own $\gamma_{\rm eff}(x_{\rm H_2}, T)$, so the invariant is wrong wherever the H_2 fraction or the temperature varies across the boundary stencil, which is the whole molecular base. c_i is the *interior* sound speed at the interior $\gamma_{\rm eff}$, because the outgoing wave is an interior wave.

2.2 The reservoir is (p, s), stated at the base level

The two reservoir conditions are the pressure and the specific entropy of the lower atmosphere at the base level, not the pressure and the density. The difference matters when the base *composition* changes: with (p, s) the temperature follows the composition, with (p, T) the entropy does. The entropy is not stored as a number; it is carried as the isentrope through $(p_{\rm base}, T_{\rm base})$ at the base composition, which is what `base.inp` and the `Lower atmosphere profile:` reader state.

`input_read` calls `set_base_reservoir` once, with

$$p_{\rm res} = n_{\rm tot,bc} + \delta p_{\rm bc}, \qquad T_{\rm res} = 1\ (= T_0), \qquad \hat{n} = \frac{n_{\rm tot,bc} + \delta p_{\rm bc}}{\rho_{\rm bc}}, \qquad r_{\rm base\ level} = 1, \tag{4}$$

where $\hat{n}$ is the particle count per unit mass at the base and $\delta p_{\rm bc}$ is the placeholder 10^{-10} in force at that point of the startup. The level is $r = 1$, the planet radius the handoff quotes (1 μbar in Koskinen et al. 2022, §3.1), *not* the first face:

`base_boundary` carries the reservoir state from the level to $r_{\rm edg}(0)$ along its own hydrostatic isentrope, so the level a user states does not move when the grid changes and the half-cell error the module exists to remove is not reintroduced at the level itself.

The isentropic continuation. `continue_hydrostatic_isentrope` carries a state from r_a to r_b at rest and at constant entropy, where the Bernoulli function is uniform:

$$u(T_b)+T_b \;=\; u(T_a)+T_a-\frac{\phi(r_b)-\phi(r_a)}{\hat{n}}, \qquad \rho_b=\rho_a\left(\frac{T_b}{T_a}\right)^{c_v(T_{\rm mid})}, \tag{5}$$

with u the internal energy per particle of the caloric EOS. The temperature is found by Newton with $\mathrm{d}h/\mathrm{d}T = c_v+1$, exact for the mixture; a potential drop deep enough to extrapolate through zero temperature falls back on the isothermal continuation, the $c_v\to\infty$ limit of the same relation. VALIDITY, stated at the routine: the midpoint c_v makes the density exact to third order in $\ln(T_b/T_a)$, and the routine is used only over half a cell and over the two ghost cells, where $|\ln(T_b/T_a)|$ is 6×10^{-3} on the hot-Uranus base grid and below 0.1 on any grid this code runs, so the density error is below 10^{-7} relative. It is not a general isentrope integrator over a scale height.

2.3 The face state

`characteristic_base_face_state` builds the interior state at the face from cell 1 alone, along cell 1's own hydrostatic isentrope, and *not* from the reconstruction stencil: the stencil reaches into the ghost, and a boundary condition that reads its own output is the circular dependency the module exists to break. Continuing along the hydrostatic isentrope rather than extrapolating linearly is what makes the condition *well balanced*: in a hydrostatic atmosphere at rest the interior face pressure equals the reservoir's and (3) returns $v_b = 0$ exactly, at any resolution. The velocity is continued along the steady mass flux,

$$v_i=\frac{\rho_1 v_1 r_1^2}{\rho_i\, r_{\rm edg}(0)^2}, \qquad c_i=\sqrt{\gamma_{\rm eff}(T_i)\,p_i/\rho_i}, \qquad M_i=v_i/c_i. \tag{6}$$

The pressure is a reservoir condition in every subsonic branch,

$$p_b=p_{\rm res}, \tag{7}$$

and the velocity then follows from (3), $v_b = v_i+(p_b-p_i)/(\rho_i c_i)$. The entropy is a reservoir condition only while gas enters: on reversal the entropy at the face is the interior's, advected out, and the reservoir has nothing to say about it. The contact is upwinded on that direction,

$$\rho_b=(1-w_{\rm rev})\,\rho_{\rm res}+w_{\rm rev}\,\rho\big(s_i,p_b\big), \qquad w_{\rm rev}\in\{0,1\}, \tag{8}$$

and **the choice is discrete and has no width**. A contact carrying different entropies on its two sides has direction-dependent traces; a face density between them is an interpolation and not a law of thermal contact, and the average of the two isentropes that a symmetric handover returns at zero has no physics behind it. This is what boundary model v2 changed from v1, whose entropy source was handed over by a cubic smoothstep in the face Mach number.

The order of the three steps is what keeps the choice from being circular. The acoustic matching comes first, so v_b is a result of (3) and does not depend on ρ_b; the direction of the contact is read off the matched state next; the contact

is upwinded last. The direction is read from the mass flux at the level, and the quantity that carries it is the wind window $r \geq r_{\mathrm{flux}}$, not the first interior cell's own product $\rho_1 v_1 r_1^2$ and not the matched face velocity. At a base carrying a collocated odd-even mode the cell-centered product is not a flux at all: MEASURED on the certified LHS 1140 b atomic state the Riemann face flux is $+1.00000\,F_{\mathrm{wind}}$ at every face while the cell-centered product reads $-2.00\,F_{\mathrm{wind}}$ at cell 1, and the hot-Uranus base reads $-196\,F_{\mathrm{wind}}$ against a face flux of +0.98. The matched face velocity carries $(p_{\mathrm{res}} - p_i)/(\rho_i c_i)$, and a converged state leaves that pressure difference at about 10^{-4} of p, which at a base Mach number of 3×10^{-7} is a velocity two orders above the flow's own. The window has standing where it carries a flux at all ($|M_{\mathrm{wind}}| \geq 10^{-12}$) and carries one flux (its relative standard deviation over the physical cells of the window at or below 2×10^{-3}); otherwise the matched face velocity decides, with a velocity below the rounding of the cancellation it is formed from, $4\epsilon[(|p_i| + |p_b|)/(\rho_i c_i) + |v_i|]$, counted as zero and the level given to the reservoir. The spread threshold is the geometric middle of the measured gap: 302 certified stationary states run from 4.06×10^{-5} to 1.82×10^{-4}, and the tightest state that is not a wind reads 1.39×10^{-2}.

At rest and at inflow the reservoir owns the level, and the physical assumption behind that is stated rather than carried implicitly: the lower atmosphere below the base level is dense and radiatively controlled, so on the time scales of a stationary solution it is a heat bath and a column standing at rest above it takes the level's own temperature by radiation and conduction. Its validity is a subsonic base at a level inside the radiatively controlled lower atmosphere; it says nothing about a supersonic base, and nothing about a level above that region. The one branch that does carry a declared width is the supersonic-outflow transition at $M_i = -1$, where the number of outgoing characteristics changes from two to three: a C^1 cubic smoothstep of half width `base_face_mach_blend` $= 10^{-8}$, which acts on the whole face state and never on the entropy source.

Why the reservoir holds the static pressure. The module carries a second admissible pair, in which the reservoir states its incoming acoustic invariant at rest, $p_b + \rho c v_b = p_{\mathrm{res}}$, behind the constant `base_incoming_invariant_weight`. Its value is 0 and the alternative is recorded because it was measured and rejected, not because it is a tuning knob. Both forms are exactly well balanced at rest, where $v_i = 0$ makes them identical; they differ on a moving solution, and the difference is not small (READ, module header):

	weight 0	weight 1
$\lvert R\rvert$, acoustic reflection	0.956	0.0024
$R_{\mathrm{mom}}(1)/\rho g$, at rest	3.53e-4	1.75e-4
$R_{\mathrm{mom}}(1)/\rho g$, steady wind	1.97e-4	1.32e-1

The last row decides it. Driving an exact steady isentropic inflow through the boundary (the manufactured solution of the design note's test C, base Mach 10^{-3}), weight 0 leaves cell 1 at 0.68 of the interior truncation error, so the boundary is as accurate as the scheme it bounds, while weight 1 leaves it 460 times above it: $p_b + \rho c v_b = p_{\mathrm{res}}$ makes the static pressure at the face fall below the reservoir's by a fraction M of the pressure where a real reservoir's Bernoulli drop is a fraction M^2, so it over-states the drop by $1/M$ and turns the wind's own mass flux into a base pressure error, which the momentum residual multiplies by $H/\Delta r$.

The reflection that weight 0 leaves standing is not a defect of the pair: it is what any reservoir holding a *static* quantity does to sound (a density reservoir gives the same $|R|$), and the ghost closure this replaced measured 0.953. Making

the boundary transparent to sound without corrupting the steady state needs a *relaxation* form, in which the incoming amplitude is driven toward the reservoir at a finite rate; that form is not a function of the instantaneous state, so it has no fixed point the steady residual can express. It is an open item, not a setting.

The supersonic branches, and the two guards. Supersonic *outflow* takes every characteristic out of the domain, so nothing may be prescribed and the whole face state is handed over to the interior's by the same smoothstep, evaluated at $(M_i+1)/10^{-6}$. Supersonic *inflow* would take all three characteristics inward and oblige the reservoir to state three conditions; a (p,s) reservoir states two, and this model has no lower-atmosphere velocity to state a third, so no velocity is invented here. Instead `check_base_inflow_is_subsonic` (`Apply_BC.f90`) measures $|v|/c$ on ghost 0 with the code's own adiabatic sound speed, the same one `eval_dt` uses for the CFL condition, and warns at the first step at which the boundary admits supersonic inflow, saying that the interior is then not a solution of the stated problem. It is reported and not enforced: the crossing is evidence about the configuration, not a numerical fault to repair in place.
A compatibility relation is a linear relation and, far from a solution, can ask for any velocity at all, so the face velocity is capped at `base_face_mach_max = 5` and the events counted. Four counters are kept and reported at the end of a run: evaluations at which the first interior cell was not an admissible gas state and the reservoir alone was the whole boundary (`n_base_interior_inadmissible`), Mach-capped face states, reversal evaluations, and supersonic-outflow evaluations. An inadmissible cell 1 cannot state an outgoing characteristic and dividing by it would make the boundary itself the source of a `NaN`, so `reservoir_face_state` (the reservoir carried to the face, at rest) is then the whole boundary and the run's own positivity detector reports the cell.

2.4 The ghost cell averages

The ghost is not a copy of the face. `base_ghost_averages` returns the volume average of the hydrostatic isentrope through the face state, continued *below* it, with the velocity carrying a constant $\rho v r^2$ so the reconstruction sees no spurious velocity gradient at the boundary:

$$\rho_j = \frac{\int_{r_{j-1/2}}^{r_{j+1/2}} \rho(r)\, r^2\, \mathrm{d}r}{\int_{r_{j-1/2}}^{r_{j+1/2}} r^2\, \mathrm{d}r}, \qquad v_j = \frac{\rho_b v_b\, r_{\mathrm{edg}}(0)^2}{\rho_j\, r_j^2}, \qquad j = 1-N_g, \ldots, 0, \tag{9}$$

and the same for the pressure. Writing the face value into a cell average instead is the half-cell error above: the two differ by $\Delta r/(2H)$, which is first order in Δr in the state and *zeroth* order in the momentum residual it produces. The quadrature is eight-point Gauss–Legendre, volume weighted, far beyond the reconstruction's order and costing two cells per boundary call. The routine also returns the point state at $r_{\mathrm{edg}}(1-N_g)$, the face below the ghost, which `Rec_BC` puts on *both* sides of that face so it is jump free; its flux lies outside the cells the update writes.
`Rec_BC` puts `base_face_W` straight into the left slot of the base face and leaves the right slot, the interior's reconstruction of cell 1 down to the face, alone: the Riemann solver is what mediates the two. The stored face and ghost states are a function of the interior state alone, so whoever replaces the interior owes the next `Reconstruct` an `Apply_BC`; the one path that replaces the interior without a flux update of its own is the restore of the attempted-step checkpoint, and `rebuild_state_from_checkpoint` calls `Apply_BC` for exactly this reason (§5.5).

2.5 The ghost composition: the seed and the fixed point (model v3)

The boundary is built on the two lower ghost cells, and their composition is not prescribed: it is solved by the ionization sweep like any other cell's. That makes the sweep a map from the composition it is entered at to the one it returns, because the entry state sets the columns, the cell depths, the electron and particle counts, the secondary-ionization partition, the shielding and the ionization the base handoff's partition is stated against. A ghost that is one application of that map is therefore a boundary that depends on the composition the run happened to be entered at.

It was measured, and it was not small. On one frozen physical column, reservoir, spectrum, option set and executable, two entry compositions returned ghosts 4.7×10^{-3} (H II, He II) to 5.5×10^{-2} (H_2^+) apart in their trace ions while p, ρ, T and the particle count agreed to 10^{-14} to 10^{-10}. Both were accepted by every test the sweep applies, at reaction residuals of about 3×10^{-26} with their imposed partition closed to zero. The base continuity row of the first physical cell, a near-cancellation of two face fluxes, read 8.335×10^{-8} in one and 2.923×10^{-9} in the other, against the same tolerance 7.5×10^{-9}: one certified and one refused. The difference is not unresolved iteration. Over six decades of the cell solver's stopping accuracy the two states do not approach each other at all, each staying put to 10^{-15}; neither is a fixed point, since a solve seeded with the refused ghost in that ghost's own column leaves it; and the seed-to-ghost map has a measured gain, seeds 9.7×10^{-1} apart returning ghosts 4.7×10^{-3} apart. A converged root does not remember its start.

The seed is part of the model. It is stated at the boundary and not left to the loader's handling of rows it drops: the reservoir row at the first sweep of a run, and the ghost the previous sweep returned at every sweep after it. The first is the only seed that is a function of the physical column and the declared inputs alone, which is why `load_IC` installs it in place of the ghost rows a restart pair carries; the second is the state's own ghost. The two are far apart as seeds, the reservoir row carrying 31 times the ghost's own H II and 26 times less H_3^+.

The returned ghost is a fixed point of the map that produced it. `ioniz_eq` applies its own map again at the ghost it returned, restoring the cells above the ghosts to the composition they were handed at each application, until both legs close: the largest absolute change of any species density of the two lower ghost cells, in the cell's own units, at most 10^{-11}, and the relative change of the ghost's heavy-particle plus electron count at most 10^{-6}. A boundary that cannot reach that in twenty applications stops the run by name, with both moves and the accuracy, exactly as the ghost's imposed H_2 partition closure does. What the routine returns for the physical cells is still one application of one map, so the stationary residual is still one application of one operator, and a ghost already at its fixed point, which is the ordinary case, costs one application.

The accuracy is set by the base row, not chosen: a ghost difference of 4.7×10^{-3} takes that row from 3.90 to 111.3 of its own rounding floors, so one floor of the row is bought by 4.4×10^{-5} of ghost composition and 10^{-11} is a quarter of one floor. **The norm is absolute and not per species**, because the sweep pins the composition absolutely, in the cell's own units, and not stage by stage, and no quantity the boundary builds reads a trace stage against its own value. MEASURED on a molecular carrier configuration, a ghost settled into an exact two-cycle whose H I moved by 6.5×10^{-12} and whose H_3^+ moved by 2.2×10^{-12} of the cell's composition, worth 0.06 of one rounding floor of the base row, while against their

own values those same moves read 4.1×10^{-9} and 5.1×10^{-3}; a relative statement cannot be made for a stage sitting at 4×10^{-10} of the cell. Species below 10^{-18} of the cell's own density are left out. The bracket is stated rather than tuned: 10^{-11} sits a factor 1.5 above the lowest plateau measured, so a configuration whose composition solve plateaus higher stops the run, and the answer there is to measure that plateau.

The temperature leg is inside the contract. The ghost's heavy-particle plus electron count is what turns its temperature into the pressure `base_ghost_averages` builds and into the internal energy and sound speed the caloric map returns, so a ghost whose composition has stopped moving but whose count has not is a ghost whose rates belong to one gas and whose thermodynamic state to another; the count is measured across every application and the temperature-dependent rates are rebuilt from the composition each application is entered at. Because the applications are *one* sweep, nothing that separates one sweep from the next may turn over between them: the frozen background's readiness is held at the value the sweep was entered with and takes the sweep's own value only when the sweep ends.

The contract covers the lower ghosts of a run whose network is molecular and carries helium, which is where the boundary's composition holds the trace ions the base row reads. A run outside it is marked reached before the first traversal and leaves after it, so not one number of such a run moves; the atomic ghost is not under the contract and its seed dependence is not measured. On the state that motivated it, three seeds 9.7×10^{-1} apart now certify with the same cell-1 row 2.923×10^{-9} and stand at most 8.0×10^{-11} apart over every column of both state files.

The model identity, and an older one. The two statements above are what the identity `characteristic_face_ps_reservoir_C_mi ghost_fixed_point_seed_reservoir_row_v3` adds to v2. It is written into every state file, together with the prescribed reservoir and the version of its prescription. A state written under an older model is *reported* and loaded, not refused: `load_IC` names the model that wrote it, names the model this run solves, and says that the boundary is rebuilt from the physical column and this run's reservoir and that its ghost composition is solved to this model's own seed and fixed point, so the state's residual is not the residual it was written with. A file written before the identity was recorded says that instead. The seed of record, whether a diagnostic has replaced it, and what the fixed point reached are all printed in the boundary report.

2.6 What fixes the reservoir pressure

The reservoir pressure is $n_{\mathrm{tot,bc}} + \delta p_{\mathrm{bc}}$, with $n_{\mathrm{tot,bc}} = (1 + Y + \sum_X a_X)/(1+Y)$ the total nuclei density at the base in units of n_0 (= 1 for H/He only, reduced by the H_2-bound share under a molecular base) and δp_{bc} the base electron density in units of n_0,

$$\delta p_{\mathrm{bc}} = \left. \frac{n_{\mathrm{HII}} + n_{\mathrm{HeII}} + 2\,n_{\mathrm{HeIII}} + \sum_i q_i\, n_i^{\mathrm{met}}}{n_0} \right|_{j = 1 - N_g} = \frac{n_e(\mathrm{base})}{n_0}, \tag{10}$$

recomputed after every ionization-equilibrium solve (`ionization_equilibrium.f90`), with the metal charges q_i included under `eos_metals 1`. Through the EOS closure (2) the pair (nuclei density $n_{\mathrm{tot,bc}}$, pressure $n_{\mathrm{tot,bc}} + \delta p_{\mathrm{bc}}$) is the statement $T = T_0$: as photoionization develops at the base and free electrons add partial pressure, the pressure is raised by exactly the electron share so the base is not artificially cooled. For a near-neutral base, which is the usual situation since the base optical depth is huge, δp_{bc} is tiny.

The reservoir is resolved once, at startup. `set_base_reservoir` is called from `input_read` and from nowhere else in the production path, so the reservoir the boundary states is the startup pair, with $\delta p_{\rm bc} = 10^{-10}$ and the $n_{\rm tot,bc}$ in force at that moment. The later refresh of $\delta p_{\rm bc}$ by the ionization sweep, and the molecular refresh $n_{\rm tot,bc} = n_{\rm tot}(1-N_g)/n_0$ that the sweep applies when the molecular chemistry is on, therefore reach $n_{\rm tot,bc}$ and $\delta p_{\rm bc}$ themselves but not the reservoir built from them. For a near-neutral base the two agree to $\delta p_{\rm bc}$; the discrepancy is the molecular case, and it is recorded here as read from the source rather than measured.

2.7 Composition at the base: one source

$\rho_{\rm bc}$, $m_{\rm gas}/{\rm H}$ (`mass_per_H`) and $n_{\rm tot,bc}$ are produced by the three functions `comp_rho_bc`, `comp_mass_per_H` and `comp_ntot_bc` of `src/modules/functions/composition.f90`, which `input_read` calls once; that module is the single place where the base composition policy lives, so no code path can disagree with another about what the base is made of. The dimensionless base density is

$$\rho_{\rm bc} = \frac{m_{\rm gas}/{\rm H}}{1+Y}, \qquad m_{\rm gas}/{\rm H} = 1+4Y+\sum_X a_X A_X, \qquad a_X \equiv \frac{n_X}{n_{\rm H}}, \tag{11}$$

the mean gas mass per H/He nucleus in units of $m_{\rm H}$. Because mass densities are normalized by $n_0 m_{\rm H}$, this states "H/He nuclei density = n_0 with the prescribed composition": the hydrogen nuclei density at the base is $n_{\rm H} = n_0/(1+Y)$, the helium density Y times that, and each metal $a_X n_{\rm H}$. The metal mass term $\sum_X a_X A_X \approx 0.016$ at solar abundances (a $\sim$1.2% effect) is included by the default `eos_metals 1` policy; `eos_metals 0` restores the legacy H/He-only trace approximation, $\rho_{\rm bc} = (1+4Y)/(1+Y)$, which is therefore the value of a metals-off atomic base and not the general form. Four switches move the numbers inside the module:

- `eos_metals` (`metals.inp`; default 1 since 2026-06-13) puts the trace-metal mass into $m_{\rm gas}/{\rm H}$ and the trace-metal nuclei into $n_{\rm tot,bc}$. With `eos_metals 0`, or with no metals, both collapse to the H/He values $1+4Y$ and 1.

- `Molecular base: True` (Tier-2a; auto-enabled whenever the molecular chemistry is on) subtracts from $n_{\rm tot,bc}$ the particles lost to the H nuclei bound into H_2 at the base: two H nuclei make one molecule, so a bound H fraction x_2 costs $x_2/2$ particles (`h2_bound_fraction`). This changes the particle count only: $\rho_{\rm bc}$, hence the mass reservoir, is untouched, and the species arrays stay atomic.

- A `base.inp` handoff supplies $q_{\rm H_2}$ at the base (`q_H2_base`) and then takes precedence over the chemical-equilibrium fit that `h2_mixing_ratio_base` otherwise evaluates at $(p_{\rm base}, T_0)$; `input_read` echoes which of the two set $n_{\rm tot,bc}$. The same handoff can move the *reservoirs* these formulas are built from, not only the particle count: `HeH_base` sets Y, and one `<El>_H_base` key per element sets the corresponding a_X, overriding `metals.inp` for that element and turning the metal system on by itself when it is the only nonzero abundance. `thereis_metals`, `melem_ab` and `thereis_lowIP_metal` are all derived *after* the handoff, so a handoff element enters $\rho_{\rm bc}$, $m_{\rm gas}/{\rm H}$ and $n_{\rm tot,bc}$ exactly as a `metals.inp` element does.

- A lower-atmosphere *profile* (`Lower atmosphere profile:`, `lower_atmosphere_profile.f90`) states T_0, R_0, $p_{\rm base}$, $q_{\rm H_2}$, Y and every elemental ratio it carries at its own matching pressure, and is read before `read_base_inp`

and applied after it, so nothing scalar can overwrite a profile value. With one in use the EOS-boundary, elemental-reservoir and boundary-constraint keys of `base.inp` are refused by name. The columns `n_tot` and `rho` of the profile are carried but *not* imposed: the base density stays the n_0 of `input.inp` and $n_{\mathrm{tot,bc}}$ still follows from T_0, Y, $q_{\mathrm{H_2}}$ and the elemental ratios through `comp_ntot_bc`, exactly as on the scalar path. Schema: `md/input_schema.md` §2d.

2.8 The keys that no longer exist

Four input keys named components of the ghost-cell closure the characteristic condition replaced, and each is now an *error* rather than a warning (`retired_base_key`, `input_read.f90`): a ghost velocity copied from the interior and smoothed (`Valve eps`), a ghost pressure extrapolated from the interior gradient (`Hydrostatic base`), a ghost temperature copied from cell 1 (`Base ghost temperature: continuous`) and a ghost velocity taken from the wind's flux constant (`Base velocity: massflux`). None has a meaning in a boundary condition that states a reservoir at the face and takes one relation from the interior, and a run that silently ignored one would answer a question the input did not ask, so it stops. The base *level* is unaffected: `Base BC: density|pressure` still states where the boundary is, and it is the key that carries over.

For the same reason the state files no longer report `valve` and `fluxconst` in their coupling header, and the `base_bc` field of the provenance line names the boundary (`characteristic`) rather than selecting among options; the field is kept in its position so that a reader of an older profile can still tell which boundary produced it (`utilities.f90`, `write_provenance_header`).

2.9 The "breathing base"

The base cells of a marching run do not settle onto an exact $v = 0$ fixed point but onto a small limit cycle around it: the wind draws mass up, the column over-supplied against what the heating profile above can carry sinks back under gravity, the returning gas raises ρ and p in the first few interior cells, and the accumulated pressure pushes it out again. This is the *breathing base*. It was present in the original ATES and under the retired valve closure, where the asymmetry of the boundary (an open pipe outward, a wall inward) drove it; the characteristic condition has no valve, and what remains is the $\mathrm{O}(\Delta t)$ kick of a marching start together with a limit cycle that switches off below a CFL of about 0.15–0.3, with the steady state at its root (`docs/Update_EXHALE_stage1.pdf` §161; READ, not re-measured here).

Two regimes are still worth distinguishing (see `docs/initial_condition_benchmark.pdf`, §7):

Self-limiting (normal). The pressure built up during the inward phase immediately produces the restoring force, so the amplitude saturates. The oscillation stays confined to the first few cells ($r \lesssim 1.1\,R_p$) and has no measurable effect on the wind region or on $\dot{M}$.

Self-amplifying (pathological). If the *global* initial condition is inconsistent (the canonical example: a naive full-Parker density profile, whose hot column drains onto the base), each cycle deposits more mass than the previous one and the breathing grows until the run diverges. The cure is a consistent IC (hydrostatic density, optionally with a warm seed overlay), not a change of the BC.

Numerical consequences. The oscillation interacts with three parts of the code:

- *Convergence monitoring.* The breathing cells never stop oscillating, so a convergence criterion of the form $\max_j |\Delta u|$ over the whole grid can stall at the 10^{-2}–10^{-1} level indefinitely. The momentum-flatness and *du/dtu* diagnostics are therefore evaluated only over the wind region $[\, j_{\min} : N \,]$, where $j_{\min}$ is the first cell with $r \geq r_{\rm esc}$ (the `Escape radius` input; §4.2); the layer below that window is measured and printed separately rather than accepted silently. A periodic Shapiro (1970) low-pass filter, the stabilizer CETIMB (Koskinen et al. 2013a) applies to exactly this gravity-unbalanced sound-wave instability, damps the oscillatory component (§2.11), but it biases the momentum and pushes clean cold-IC cases off the wind branch into infall, so it is kept opt-in rather than a default.

- *Newton solver.* The kink the smooth valve once existed to remove is gone with the valve. What remains as a designed-in differentiability concern is the reversal branch of §2.3, which is why the branch handover is a C^1 smoothstep of width 10^{-6} in the face Mach number rather than an `if`.

- *Advection post-processing.* The `_adv` correction integrates ionization states along streamlines; where $v \leq 0$ at the base the streamline of a base cell exits the domain through the lower boundary and the correction is ill-defined (this produced the historical temperature spike and sawtooth at $r \approx 1.1$). The post-processor keeps the equilibrium solution and skips the advection correction wherever $v \leq 0$ at the base.

2.10 Base grid resolution

Not every alternating pattern at the base is the breathing limit cycle. On the `Mixed` grid the base region is $N_{\rm low}$ uniform cells of width $\Delta r_{\rm base}$ stacked on $r = 1$, and the physical requirement on that width is that it resolve the base density scale height $H = kT/(\mu g)$. Where $H/\Delta r_{\rm base}$ is of order a few cells, the discretization supports a *stationary* $2\Delta r$ entropy (contact) mode that nothing in the scheme damps: HLLC resolves a contact of zero speed exactly, the gravity source is cell-local, and the WENO3 pressure gradient sees only interface pressures. The measured dependence is an alternating amplitude in $\ln\rho$ of $\sim 10^{-2}$ for $H/\Delta r < 5$ against $\sim 10^{-4}$ for $H/\Delta r > 100$. High-gravity planets have the least margin: on the default grid `write_setup_report` echoes 26.9 cells per H at $T_{\rm eq}$ for HD 189733 b against 102.3 for WASP-121 b. The resolution is therefore an input key, `Base grid [dr,cells]: <dr_base> [<N_low_cells>]` (`input_read.f90`; defaults `2.0e-4 50`, ignored by the `Uniform` and `Stretched` grid types). The two numbers are not independent (their product is the radial extent of the uniform region, $0.01\,R_p$ with the defaults), so refining the base at fixed extent means dividing $\Delta r_{\rm base}$ and multiplying $N_{\rm low}$ by the same factor. Because those extra base cells are taken out of the same total N, the runtime `Grid cells:` key of §1. is what lets a refinement keep the outer stretched region unchanged; the paired settings are

`Base grid [dr,cells]:`	`Grid cells:`
`1.0e-4 100`	500
`5.0e-5 200`	658
`2.5e-5 400`	916
`1.25e-5 800`	1375

The default width is the double 2×10^{-4}, so spelling it out (`Base grid [dr,cells]: 2.0e-4 50`) builds the same grid as leaving the key out. Runs made before 2026-09-19 without the key used the single-precision neighbor $1.9999999494757503 \times 10^{-4}$, 2.5×10^{-8} relative below it, which moves the cell centers by 3.8–6.6 $\times 10^{-9}$ relative on the grids measured, enough for `load_IC` (tolerance 10^{-10}) to refuse their states on the present default grid. Every such input that has stored results beside it carries the line `Base grid [dr,cells]: 1.9999999494757503e-4 50`, which rebuilds that grid to the bit.

2.11 The three remaining opt-in controls

Three keys still act on the base region, and all three default to off or to the legacy setting, so a run that does not ask for them is unchanged. The measured effects of the base anchor and of the Shapiro filter, taken under the retired ghost closure, are tabulated in §7..

Base anchor (`Base BC: density | pressure [<p_ubar>]`). Which quantity states the base level. `density` (default, legacy) fixes the base number density at $n_0 = 10^{\texttt{Log10 lower boundary number density}}$, and the reservoir of §2.2 is the isentrope through ($p = n_{\mathrm{tot,bc}} + \delta p_{\mathrm{bc}}$, $T = T_0$) at that density. `pressure` (CETIMB-style) derives n_0 instead, so that the base pressure $n_0 k_B T_0 n_{\mathrm{tot,bc}}$ equals `p_ubar` microbar (default 1 μbar, the CETIMB lower boundary); the reservoir pressure is then that value and its temperature T_0, both stated at the base *level* $r = 1$ and carried to the first face by the isentrope, so the level does not move with the grid. The physical effect is a much less dense base (1 μbar is about 20$\times$ below $n_0 = 10^{14}$), which shrinks the dense-base gravity source ρg.

Shapiro filter (`Shapiro filter: <eps> [<every>]`). Every `every` steps a weak 1–2–1 low-pass filter $u_j \mathrel{+}= \frac{\varepsilon}{4}(u_{j-1} - 2u_j + u_{j+1})$ is applied to the conservative variables (`shapiro_filter`, `Apply_BC.f90`; a no-op at $\varepsilon \le 0$, the default). This is CETIMB's cure for the gravity-unbalanced sound waves and it does smooth the transient. On otherwise-clean cold-IC cases it injects a systematic momentum bias that drives the solution off the transonic-wind saddle into the infall branch ($v < 0$ almost everywhere; the `_Shap` rows of Table 3), so it is useful only for cases that genuinely breathe. It touches the marching state alone, which leaves the Newton solver solving a system whose mode the filter had been hiding.

Low-Mach contact-mode damping (`Low-Mach damping: <eps4> [<M_th>]`). Not a boundary condition, but aimed at the same near-base region. The HLLC flux carries the contact/entropy wave at $S_* \sim v$, so the dissipation it applies to that family vanishes as the flow stagnates and a $2\Delta r$ mode in v and T becomes marginally damped, which is what a shell where metal cooling has stopped the flow looks like. The key adds a gated fourth-difference (Jameson–Schmidt–Turkel) stress to the numerical *momentum* flux, together with its work term $v D_p$ in the energy flux so the pair conserves total energy; the mass flux is untouched, and the gate closes the term identically above $M > M_{\mathrm{th}}$ (default 10^{-3}). It lives in `src/modules/flux/low_mach_dissipation.f90` and is applied inside `RK_rhs`, the one routine the steady residual assembly also calls, so the marching loop and the Newton residual see the *same* equation. Off by default ($\varepsilon_4 \le 0$ skips the code path entirely). *Caveats, both measured*: the coefficient has an operating window, `input_read` warns above the explicit stability bound $\varepsilon_4 < 1/(16\,\mathrm{CFL})$, on the `examples/16` warm restart only 5×10^{-3} and 10^{-2} gave a clean Newton finish, and on the *cold* start of that same configuration no coefficient helped at all.

Implementation note. The cell-by-cell ionization-equilibrium sweep is OpenMP-parallel over cells (the metal and charge-exchange coefficients in each cell are kept thread-private); the parallel result is bit-identical to the serial one, so none of the BC/IC behavior below is affected.

2.12 Temperature, ionization, and metals at the lower boundary

No boundary condition is imposed on the temperature or on any ionization fraction. The ghost cells are included as ordinary cells in the ionization-equilibrium sweep (`ionization_equilibrium.f90` loops over $j = N{+}N_g, \ldots, 1{-}N_g$), so their T, H/He stages, He 2^3S, and all metal stages come out of the same coupled MINPACK solve as the interior, evaluated at the ghost state the characteristic condition wrote (§2.4). Combined with §2.6 this makes the effective thermodynamic boundary condition "the reservoir isentrope through (n_0, T_0), ionization in local equilibrium". Metal abundances enter only through the runtime `metals.inp` fractions; the boundary needs no special metal treatment: their abundances ride on the reservoir composition, and under the default `eos_metals 1` policy their mass, electrons, and nuclei are carried consistently through `calc_rho`/`calc_ne`/`calc_ntot` (`eos_metals 0` restores the legacy trace approximation in which they never appear in the mass or momentum budget).

2.13 Boundaries of the transported composition columns

The hydrodynamic ghosts above are not the only boundary the code states. When an element mass fraction or a molecular carrier is *transported*, its column carries its own boundary, and in the stationary solve those ghosts must belong to the iterate being evaluated rather than to whatever the last sweep left behind.

At the top: the iterate's own zero gradient. The element and carrier operators pose a zero-gradient outflow condition at the top of the column: nothing enters the domain from outside through an element, the diffusive face coefficients stop at face $N-1$ so the diffusive flux vanishes at face N, and with the ghost carrying the outermost cell's composition the advective reconstruction sees no compositional step across the top. `write_species_rows_into_composition` (`steady_newton.f90`), the one place a trial composition is assembled for the residual, therefore writes the upper ghosts of the transported element and carrier columns from the outermost cell of *that trial*, by the same rule the marching path uses ($Y(N{+}1{:}N{+}N_g) = Y(N)$ in the advective stage of `binary_element_diffusion.f90`, $f_c(N{+}1{:}N{+}N_g) = f_c(N)$ in the carrier operator). With a zero-gradient element mass fraction the element mixing ratio is zero-gradient too, since $m_{\rm sum}/n_{\rm H} = m_1 + m_{\rm He}\,(n_{\rm He}/n_{\rm H})$ is fixed by the helium mass fraction: MEASURED at 1.4×10^{-16}. Before that write the outermost element row of the element reload read 1.19×10^{-2} of its own scale against an interior level of 1.3×10^{-13}, and the sodium row of the outermost cell bound the solve on 51 of 109 iterations; with it, on 0 of 252.

The operator itself takes the caller's ghosts. `element_transport_residual` writes no ghost of its own: it is measured under the boundary its caller states, while its relaxation twins (`element_diffusion_step`, `solve_trace_element_in_hydrogen`) refill their own working arrays before every pass. That asymmetry is deliberate and it is documented at the routine. A refill was implemented in the residual and then reverted: it changes the residual by one element's outermost row in its twelfth digit and nothing else, yet the atomic element rows do not survive a last-bit change of the upper ghosts, whatever makes it. MEASURED: a control that merely multiplies the

same three ghost quantities by $1+2\times10^{-16}$, posing no boundary at all, moves $\|R\|$ at iteration 40 from 3.212×10^{-2} (the caller's ghosts) to 1.125, against 1.199 for the refill. The reading that matters beyond this item is that $\|R\|$ at a fixed iteration count is not a reproducible quantity for that solve; the size of the caller dependence (1.19×10^{-2} on a stale caller) is therefore asserted rather than assumed away.

At the base: two different equations. The lower ghosts of the transported columns are never rewritten by the iterate: they are the Dirichlet reservoir the handoff states. The equation the *first physical cell* carries differs by row class, and `species_row_base_equation` is its single definition, read by both `eval_residual` (which assembles the equation) and `cell_row_scales` (which nondimensionalizes it), so the two cannot drift apart:

- an **element row** is a Dirichlet reservoir condition, $X(1) = X_{\text{reservoir}}$. The element operator poses no transport balance in cell 1; that cell *is* the reservoir the column is anchored to, and the scale that makes the equation dimensionless is the unknown's own.
- a **carrier row** is a control-volume balance, as in every other cell. The inner ghost is data and not an unknown: it carries the composition of the handoff, and where the base face flows inward the advective term it contributes is a constant of the row. The cell balances transport against chemistry, so the scale is the row's own terms.

The projection back onto species conserves mass. Writing element mass fractions back into a species vector goes through `project_elements`, and what it must preserve is the mass the species handed in carried. The mass is read from the cell itself (`mixture_mass_split`: the trace metals at the abundance the cell holds, not at the reservoir's) and component 1, the hydrogen carriers together with the trace metals and the heavy nuclei bound in the molecular carriers, is scaled as a whole. Taking the mixture mass with the reservoir mass per hydrogen nucleus instead wrote the difference into hydrogen at every call: MEASURED at one cell of the certification column, with metal/H 0.024401 against the reservoir's 0.028227, $\sum_i m_i f_i$ went $1.0000000 \to 1.0000636$ after one step, helium's gain exceeding the hydrogen and metal loss, which is mass created. After the fix the closure of the certification column stands at 2.3×10^{-15} after the relaxation (from 3.37×10^{-3}) and the handed-back atomic candidate at 1.94×10^{-9}, the file's own precision. This is also what makes the writer-to-loader round trip return the state (§5.5).

3. Upper boundary: free outflow

At an outflowing outer face the characteristics $v+c$ and v leave the domain; $v-c$ leaves too only where the outflow is supersonic. Where it is subsonic, $v-c$ enters and one condition has to be supplied from outside; the ghost, a continuation of the interior, supplies it implicitly, so it can then set the mass flux, and the outer face is placed past the sonic point where a run can place it (update log stage 3 sections 127.4 and 127.6). The rule is stated once, in `free_outflow_ghost` (`Apply_BC.f90`), for every reconstruction: which characteristics cross a face is a property of the flow, not of the scheme that reads the ghost. For a subsonic outermost cell the continuation is the isothermal hydrostatic one,

$$\rho_g = \rho_N \exp\left[-\frac{\phi_g-\phi_N}{p_N/\rho_N}\right], \qquad p_g = p_N \exp\left[-\frac{\phi_g-\phi_N}{p_N/\rho_N}\right], \qquad v_g = v_N, \tag{12}$$

the state of rest the last physical cell carries outward under the code's own potential, with p/ρ, the isothermal sound speed squared of the mixture, held at cell N's value. Since ρ and p are scaled by the same factor, $p_g/\rho_g = p_N/\rho_N$: the ghost is at the *same temperature* as cell N and differs from it only in density, so everything downstream that reads a ghost (the ionization sweep, the sound speed of `eval_dt`) sees the temperature it saw before, at the density the stratification gives. The absorbing columns do not read the ghosts (§4.).

A supersonic outermost cell: the steady-wind ghost. Where v_N exceeds the adiabatic sound speed of cell N, the ghost is the gas the wind carries beyond the face, not a state at rest, and the hydrostatic continuation is outside its validity condition ($M^2 H/L_v \ll 1$). Past the ridge of the Roche potential (L1) it is also unbounded: ϕ falls outward and the factor above exceeds one. MEASURED with the binary of before 2026-10-11: WASP-121 b at $12\,R_p$, step 0, $1.7 \times 10^6\,\mathrm{cm}^{-3}$ in cell N against 3.4×10^{34} and 4.9×10^{63} in the ghosts; WASP-52 b at $10\,R_p$, step 6780, 3.3×10^3 against 3.6×10^{22} and 8.1×10^{41}, after which the energy update failed. The ghost of a supersonic cell N is therefore the steady isothermal wind through it at $c^2 = p_N/\rho_N$:

$$\tfrac{1}{2}v_g^2 - c^2 \ln v_g = \tfrac{1}{2}v_N^2 - c^2 \ln v_N + 2c^2 \ln \frac{r_g}{r_N} - (\phi_g - \phi_N), \qquad \rho_g = \rho_N \frac{v_N r_N^2}{v_g r_g^2}, \qquad \frac{p_g}{\rho_g} = c^2, \tag{13}$$

with v_g the supersonic root (`isothermal_wind_speed` in `grav_func`, the solver of the transonic IC) and $v_g = c$ where a steep rise of ϕ leaves no supersonic root. The mass flux is constant through the ghosts in any potential. The face is supersonic, so the ghost enters only cell N's reconstruction slope.

Why a gradient and not a copy. A zero-gradient ghost asserts $\mathrm{d}p/\mathrm{d}r = 0$ at the outer face, which is a hydrostatic imbalance of the order of the local weight ρg that no resolution removes. Under PLM it is worse: the forward difference entering the MC limiter is then exactly zero, so the limited slope of cell N is exactly zero and that cell carries no pressure gradient at all. MEASURED on the analytic hydrostatic column, the momentum residual of cell N in units of its own weight at `Grid cells` $= 250/500/1000/2000$ and its order in $1/N$ (READ from the routine's recorded measurement):

Outer ghost rule	250	500	1000	2000	order
zero-gradient copy (PLM)	0.759	0.753	0.751	0.750	0.006
linear extrapolation	0.059	0.021	0.008	0.0030	1.42
the continuation above	0.0044	0.0011	0.00032	0.00010	1.80

With the continuation the largest residual of the domain is no longer at the boundary but at the base/stretched grid junction, where the interior truncation error is.

The velocity stays a copy. The outgoing characteristics carry the velocity out of the domain and the boundary has nothing of its own to say about it; a copy states the least and can neither reverse the flow, which would make the boundary an inflow carrying characteristics inward that only a reservoir may state, nor manufacture a speed. Two gradient-carrying rules were measured and both rejected on `wasp_full`, whose outer edge is subsonic at Mach 0.96 so that the ghost reaches the interior: a linear extrapolation in r carries about 1 percent less mass flux than cell N and the run then sits in the base-breathing limit cycle with du between 4×10^{-3} and 1.6×10^{-2}, never reaching the 10^{-3} marching stop within 43000 steps where the copy stops at 17135; and $v_g = v_N(\rho_N/\rho_g)(r_N/r_g)^2$, the steady continuity

equation, stops the same case sooner (14442 steps) but divides by a ghost density the stratification can make arbitrarily small, so where the ghost cell is thicker than a scale height it manufactures an arbitrarily large speed at the boundary.

Positivity. The exponential of a finite argument with $p_N/\rho_N > 0$ is positive, so an admissible cell N gives an admissible ghost. A cell N that is not itself an admissible gas state, or a potential difference large enough to overflow the exponential, leaves the ghost with the zero-gradient copy of cell N and is counted in `n_ghost_cells_positivity_limited`, reported at the end of the run. The tests are written as the negation of "admissible", so a `NaN`, which compares false against everything, is caught too. A nonzero count is a statement about the interior, not about the boundary.

Where it matters, and where it does not. A supersonic outflow carries no information inward *through the Riemann solution*: with every wave speed of the face positive, HLLC returns the physical flux of the left state and never forms an expression in the right state. It does still reach cell N through that cell's own reconstruction stencil, which is a property of the stencil and not of the characteristics, so a supersonic run moves under a change of this rule by the size of cell N's limited slope and no further: MEASURED on `mol_base_handoff` at 12000 steps, whose outer edge sits at Mach 1.92, the outermost cell moves by 3.8×10^{-4} in ρ and is the only cell of 500 that moves by more than 10^{-4}, the mass flux above $r_{\rm flux}$ is unchanged to 1.9×10^{-8} and du to five digits. The continuation therefore matters where the outer edge is *subsonic*, and that is the state it is built for: there gravity is the leading term of $\mathrm{d}p/\mathrm{d}r$, the ram contribution $\rho v\,\mathrm{d}v/\mathrm{d}r$ being smaller than $\rho\,\mathrm{d}\phi/\mathrm{d}r$ by the square of the Mach number.
Two caveats about the model rather than the closure:

- If the flow is *not* supersonic at $r_{\rm max}$ (observed for deep Roche-lobe-overflow geometries, e.g. WASP-121b Case D, which is subsonic at L1), one in-going characteristic formally remains and no reservoir states it; in practice the scheme remains stable but the outer solution should be treated with caution (geometry-limited regime).
- The BC carries no information about the medium beyond $r_{\rm max}$ (stellar wind, confinement pressure); EXHALE models the planetary outflow only.

For the reconstructed interface states `Rec_BC` copies the right state outward under PLM and extrapolates it linearly under WENO3, whose wider stencil is sensitive to the artificial flattening a plain copy would imply; these are face states and are left unguarded, because `positivity_limited_faces` already drops any face that leaves $\rho > 0$, $p > 0$ back to the cell averages.
No ionization or temperature condition is applied at the top either; the ghost cells participate in the equilibrium sweep like all others, with the (ρ, p) of the continuation.

4. Boundary-adjacent settings

These are not boundary conditions of the hydrodynamic update, but they are boundary choices of the model and are easily confused with BCs.

4.1 Radiative transfer: irradiation from the top

The stellar EUV/X-ray flux enters through the *outer* boundary. The column densities that attenuate the ionizing flux are integrated top-down (`calc_column_dens`, `utilities.f90`) from the outer face $r_{\rm edg}(N)$: the outer ghost cells carry no column, and the recursion $N(j) = N(j{+}1) + n_j \Delta r_j$ starts at cell N and proceeds inward, so the optical depth is zero at the outer face and grows toward the base. The ghosts are a closure of the reconstruction stencil, not gas on the line of sight; until 2026-10-11 the recursion started at the outermost ghost, and past the Roche ridge an exploding hydrostatic ghost then shielded the whole domain (update log stage 3 section 138). This is the radiative-transfer analogue of a boundary condition: the unattenuated stellar spectrum is imposed at $r_{\rm max}$, and the base receives whatever survives the overlying column. Because the base column is huge, the lower boundary is effectively in radiative shadow: consistent with holding it at (n_0, T_0).

4.2 Escape radius and the convergence window

The `Escape radius [R_p]:` input ($r_{\rm esc}$, default 1.5) defines $j_{\rm min}$, the first cell with $r \geq r_{\rm esc}$ (`define_grid.f90`). All steady-state diagnostics (momentum-flux flatness $\rho v r^2$ = const, du, dtu) are evaluated over $[\, j_{\rm min} : N \,]$ only, deliberately excluding the breathing base (§2.9). A guard added for deep-RLOF geometries (EXHALE): if $r_{\rm esc} \geq r_{\rm max}$ the window would be empty (and `maxval`/`minval` over an empty section return $\mp$`HUGE`, producing a spurious instant "convergence"); in that case the window is clamped to the outer half of the domain and a loud warning asks the user to lower $r_{\rm esc}$.

2026-08-15: the layer below the window is measured, not accepted silently. The acceptance window is still $[\, j_{\rm min} : N \,]$, but the layer $[\, 1 : j_{\rm min}{-}1 \,]$ excluded from it is no longer unreported. `resid_relnorm_below_escape` (`src/modules/time_step/steady_newton.f90`) applies the same volume-weighted relative norm to that layer and locates the cell carrying the largest cell-wise residual in it; `write_resid_below_escape` prints one line at the start and end of the PTC solve and at the start and end of the JFNK solve. The value is printed and never tested: nothing in the solve depends on it.

The momentum row of that diagnostic is *not* scaled by $|u_2| = |\rho v|$. The layer is quasi-hydrostatic ($|v| \sim 10\,\mathrm{cm\,s^{-1}}$), so that denominator collapses and the ratio is large wherever the flow is slow, whether or not the state is right: converged states read 0.2–10 on it, the metals-off solution nobody doubts reading the largest. The physical scale of the momentum equation there is the hydrostatic pair $|\mathrm{d}p/\mathrm{d}r| \sim \rho g$, so the momentum row is divided by the gravitational force density

$$s_{\rm grav}(j) = |\rho_j| \, \frac{|G\phi_i(j) - G\phi_i(j{-}1)|}{\Delta r_j}, \tag{14}$$

the same discrete potential difference the momentum source term uses (`Source.f90`); the printed number is then the fractional violation of hydrostatic balance in the layer. On that scale the converged HD 209458 b solutions (including the metals-off one that read 9.99 before) all hold the layer at 10^{-5} to 1.4×10^{-4} of ρg, comparable to the wind-region `Resid tol`. The mass and energy rows keep the $|u|$ scale and print separately.

Widening the acceptance window to include the layer was considered and is *rejected by measurement*: on the old $|u|$ scale a converged state carries 0.2–10 there, four to six orders above `Resid tol`, so the widened window would reject solutions that are correct.

4.3 Domain truncation (where the upper boundary sits)

In Roche mode the grid ends by default at the Hill radius $r_{\rm max} = (3M_*/M_p)^{-1/3}\tilde{a} \approx$ L1, and at the user-supplied `Outer radius [R_p]`: when that key is given, with the full Roche potential kept past L1 (as Huang et al. 2023 and Yan et al. 2022 do); in spherical mode at `Outer radius`. The choice changes where the free-outflow BC is applied and whether cell N is supersonic, and hence which of the two ghost rules of §3. applies.

5. Initial conditions (`set_IC.f90`)

Three concepts to fix first: cold IC, warm IC, and starter. For the reader's convenience, every IC family below is a variation on three ideas. The distinction matters because it controls whether the run starts *inside* the wind basin or has to be pushed there.

Cold IC. The default. A fully *neutral*, isothermal ($T = T_0$) *hydrostatic* atmosphere with only a small outward velocity seed. Nothing is pre-heated or pre-ionized: the simulation grows the temperature, ionization, and wind from scratch. Because it is built from EXHALE's own base BC (fixed $\rho_{\rm bc}$, T_0), it is automatically *consistent with the lower boundary* and relaxes cleanly to the transonic wind.

Warm IC (warm seed / warm start). An initial state that already carries a *heated, ionized, flowing* profile, overlaid on the hydrostatic density: the transonic isothermal-wind and hot-Parker families. A plausible wind therefore exists from step 0, which is needed where a cold start cannot launch (deep-RLOF or low-gravity boil-off). It is still EXHALE's own base with a thermodynamic overlay, so it stays consistent with the lower boundary; it only auto-enables `force_start` so the heating can settle.

Starter (starter solution / seed). A *complete, externally pre-converged* steady wind solution read in wholesale as the IC: the Wind-AE warm start (`IC mode: windae`), which ramps a Wind-AE `windsoln` (or a grid/database seed) to this planet. Unlike a warm seed, a starter *replaces the entire state* with another code's solution, including its own base; that base can clash with EXHALE's lower BC and pull the run off the wind branch (§7.). A starter bootstraps hard cases but is not guaranteed consistent with EXHALE's boundary.

When a run does not restart from saved profiles (§5.5), the initial state is built by `set_IC` in `src/modules/init/`. Five options are available: the three IC families `set_IC` actually constructs (cold hydrostatic, transonic isothermal wind, hot-Parker warm seed), plus `IC mode: auto`, which selects among them at run time, and `IC mode: windae`, whose profile is built in `init.f90` from the Wind-AE solver. The choice is made by optional `input.inp` keys, with the cold hydrostatic atmosphere as the default:

Family	Key	Default
Cold hydrostatic + velocity seed	(none)	yes
Transonic isothermal wind	`Transonic IC: True`	off
Hot-Parker warm seed	`Hot Parker IC:` $T_{\rm wind}$ `[K]`	off
Automatic selection	`IC mode: auto`	off
Wind-AE warm start	`IC mode: windae`	off

The transonic and hot-Parker families, and only those two, force the first 1000 iterations (`force_start`) so that the heating structure can develop before the convergence test is allowed to fire. Whatever the family, the full dimensional IC is dumped to `output/IC_dump.txt` for inspection before the first step.

Automatic selection (`IC mode: auto`). `select_IC_auto` (`set_IC.f90`) picks the family from a single exact probe: `find_sonic` at the *cold* base sound speed in the actual potential. An interior sonic point selects the transonic IC; this catches both deep-RLOF bases (the Roche $\phi' \to 0$ topology toward L1 always crosses the critical condition; e.g. WASP-121b Case D, $r_c = 1.30$) and low-gravity boil-off planets ($r_c \simeq b_0/2c_0^2$ falls inside the domain for small b_0). Otherwise the cold hydrostatic default is kept (classic EUV-heated winds: HD 209458 b has $r_c \sim 54\,R_p$, far outside the domain). There is no tunable threshold; b_0 (= the escape parameter Λ) is logged as a regime diagnostic, the warm seed is never auto-selected (the benchmark showed it buys nothing when the cold start works), and the explicit keys override `auto`.

Wind-AE warm start (`IC mode: windae`). With this key, `init.f90` builds the initial condition by running the in-tree Fortran port of the Murray-Clay/Broome *Wind-AE* steady-state wind solver (`src/modules/wind_ae/`). The bridge `wae_exhale_bridge` maps the planet's EXHALE parameters (M_p, R_0, $M_\star$, a, $F_\mathrm{XUV} = J_\mathrm{XUV}$, $L_\star$, He/H, domain mode) to Wind-AE, ramps a shipped seed solution (`inputdata/windae_seed.csv`) to the planet by continuation, runs the BVP relaxation + outward integration, and writes `output/{Hydro_ioniz,Ion_species}_IC.txt` on the EXHALE grid; `load_IC` then ingests them. The whole EXHALE-input → Wind-AE solve → IC → run is a single `EXHALE.x` invocation. This is a metal-free H/He *warm start* (EXHALE re-equilibrates the ionization and any metals from the first step); finish with `Solver: Newton` for a quantitative $\dot{M}$. Best suited to seed-adjacent hot Jupiters; far-from-seed planets may fail the ramp (the run then advises `IC mode: auto`). A standalone generator `wind_ae_ic.x` (`make wind_ae_ic`) does the same out of process. Full port/validation: `md_backup/wind_ae_fortran_port_plan.md`.
The guiding principle in all three cases is *consistency with the lower boundary condition* of §2.: every IC matches $(\rho, p, v) = (\rho_\mathrm{bc}, n_\mathrm{tot,bc}{+}\delta p_\mathrm{bc}, 0)$ at the base exactly, so the boundary never has to fight the initial state. The quantitative comparison of the families is in `docs/initial_condition_benchmark.pdf`; the conclusions quoted below come from there.

5.1 Default: cold neutral hydrostatic atmosphere + velocity seed

The original ATES initial condition, kept as the EXHALE default because it is the most robust (§5.3).

Density. An isothermal hydrostatic atmosphere in the (Roche or spherical) potential, anchored to the base density BC:

$$\rho(r) = \rho_\mathrm{bc} \exp\Big[\, b_{0,\mathrm{eff}}\big(\phi(r_0) - \phi(r)\big)\Big], \tag{15}$$

with one deliberate distortion: $b_{0,\mathrm{eff}}$ starts at the physical value (1 in code units, the potential already carrying the Jeans parameter b_0) and is increased in steps of 0.2 until the density at the domain midpoint $r_{1/2} = (1 + r_\mathrm{max})/2$ falls below 10^{-8} (`set_IC.f90`). For low-gravity planets the true isothermal atmosphere would not decay enough across the domain; the loop artificially *steepens* the initial atmosphere so that the outer half of the grid starts effectively empty.

The profile is then floored at $\rho = 10^{-8}$. In a Roche domain carried past L1, ϕ in Eq. (15) is its running maximum from the base outward, $\max_{r' \le r} \phi(r')$: inside the ridge this is ϕ itself and the profile is unchanged bit for bit; past it the density is held at its ridge value (in practice the floor) instead of rising as $\exp[+|\Delta\phi|]$, which reached $5 \times 10^{43}\,\mathrm{cm}^{-3}$ at $15\,R_p$ for HD 209458 b (update log stage 3 section 138). Equation (15) with $b_{0,\mathrm{eff}} > 1$ is therefore *not* an equilibrium: it is a concentrated cold reservoir that the subsequent heating will evaporate.

Velocity. A small outward linear seed,

$$v(r) = \tfrac{1}{2}\,(r - r_0), \tag{16}$$

zero at the base and growing outward (line 108). Starting from exact rest delays the launch of the wind; the seed gives the outflow a direction without imposing any particular magnitude.

Pressure, temperature, ionization. Isothermal at the base temperature and consistent with the lower BC (lines 137–146):

$$p = (n_{\mathrm{tot,bc}} + \delta p_{\mathrm{bc}})\,\frac{\rho}{\rho_{\mathrm{bc}}}, \qquad T = 1\ (= T_0 \text{ everywhere}), \tag{17}$$

and a mostly neutral composition: $x_{\mathrm{HII}} = \delta p_{\mathrm{bc}} \sim 10^{-10}$, $x_{\mathrm{HeII}} = 10^{-10}\,Y$, HeIII and He 2^3S zero, the rest neutral. At the base cell this reproduces (ρ_{bc}, $n_{\mathrm{tot,bc}}+\delta p_{\mathrm{bc}}$, T_0) exactly.

The physical picture: *a cold, neutral, hydrostatically stratified atmosphere is exposed to the stellar EUV flux at $t = 0$ and left to evaporate*. The first ionization-equilibrium sweep deposits the heating, the upper layers warm and expand, and the wind launches self-consistently: typically within the forced-start phase.

5.2 Transonic isothermal-wind IC (EXHALE)

Motivation: deep Roche-lobe-overflow geometries (the canonical case is WASP-121b Case D, whose sonic point sits essentially at L1) launch *no* wind from the hydrostatic IC: the run stalls in a bound, breathing state. For these cases the run must start from a flowing solution. With `Transonic IC: True`, `wind_profile` (`set_IC.f90`, lines 177–233) solves the steady isothermal wind in the actual ATES potential via its Bernoulli integral:

$$\tfrac{1}{2}v^2 - c^2 \ln v - 2c^2 \ln r + \phi(r) = B, \qquad c^2 = \frac{n_{\mathrm{tot,bc}} + \delta p_{\mathrm{bc}}}{\rho_{\mathrm{bc}}}, \tag{18}$$

where c is the dimensionless isothermal sound speed of the cold base gas. The constant B is fixed at the sonic point r_c, located by bisection on the critical condition

$$\phi'(r_c) = \frac{2c^2}{r_c} \tag{19}$$

(`find_sonic`; in Roche mode $\phi' \to 0$ toward L1, so a sign change is generic). At each radius Eq. (18) has one subsonic and one supersonic root; the transonic wind takes the subsonic branch for $r < r_c$ and the supersonic branch for $r > r_c$ (robust bisection per cell, `wind_root`). The density then follows from steady mass conservation pinned to the base BC,

$$\rho v r^2 = \rho_{\mathrm{bc}}\, v(r_0)\, r_0^2. \tag{20}$$

If no interior sonic point exists (gravity dominates out to $r_{\max}$: a bound "breeze" case), the routine reports failure and `set_IC` falls back to the hydrostatic IC of §5.1 with a console notice. If the base itself is already super-critical, the flow is taken supersonic from the base.

5.3 Hot-Parker warm-seed IC (EXHALE)

This family is the product of the initial-condition benchmark (`docs/initial_condition_benchmark.pdf`, §7), whose central negative result deserves restating: **seeding the full isothermal Parker wind (its density *and* velocity) makes the run diverge.** The hot, low-density Parker column is globally inconsistent with the non-isothermal ATES equilibrium; it drains onto the base and drives the self-*amplifying* breathing mode of §2.9 (whereas the cold hydrostatic IC breathes self-limitingly and converges). The lesson: the base may not be handed an initial state whose global column it cannot support.

The robust compromise, enabled by `Hot Parker IC:` $T_{\rm wind}$ (default $T_{\rm wind} = 10^4$ K), is a *warm seed*: keep the stable cold hydrostatic *density* of §5.1 and overlay only the fields whose relaxation is slow and stiff: temperature, ionization, and a velocity head-start. The overlay is anchored to the cold base by a spatial smoothstep window (lines 114–133),

$$s(r) = \mathrm{clip}\left(\frac{r-1}{r_{\rm tr}-1}, 0, 1\right), \qquad \xi(r) = s^2(3-2s), \tag{21}$$

with $r_{\rm tr} =$ `hp_base_rtr` $= 1.3\,R_p$ (a fixed default in `parameters.f90`; it is a module variable, not a Fortran `parameter`, but no input key exposes it), so that $\xi = 0$ at the base and $\xi = 1$ for $r \geq r_{\rm tr}$. The overlaid fields are

$$\begin{aligned} T(r) &= 1 + \left(\tfrac{T_{\rm wind}}{T_0} - 1\right)\xi, & v(r) &= v_{\rm P}(r)\,s, \\ p(r) &= \rho\,\tfrac{n_{\rm tot,bc}+\delta p_{\rm bc}}{\rho_{\rm bc}}\,T\,(1+\xi), & x_{\rm HII}, x_{\rm HeII} &: \text{neutral} \to \text{ionized as } \xi, \end{aligned} \tag{22}$$

where the $(1+\xi)$ factor in the pressure is the electron partial pressure of the ionized warm gas (particle number doubles at full ionization), and $v_{\rm P}$ is the Parker velocity head-start: the transonic profile of §5.2 re-solved with the *warm, ionized* sound speed

$$c^2_{\rm warm} = \frac{n_{\rm tot,bc} + \delta p_{\rm bc}}{\rho_{\rm bc}}\,\frac{T_{\rm wind}}{T_0} \times 2, \tag{23}$$

its *density discarded* (that is the part that destabilizes), and its velocity multiplied by s so it vanishes at the base. If the warm profile has no interior sonic point, the head-start quietly degrades to the small linear seed.

At the base $\xi = 0$ gives $T = T_0$, $p = n_{\rm tot,bc} + \delta p_{\rm bc}$, $v = 0$: the lower BC of §2. exactly, with no inverted gradients for the first step to undo.

Benchmark verdict. The warm seed converges to the same $\dot{M}$ as the cold IC with no NaN/blow-up, but yields *no speedup* (−2.3% steps in the benchmark): ATES convergence is gated by the slow momentum (du) relaxation, not by the thermal/ionization relaxation the warm seed shortcuts. The cold hydrostatic IC therefore remains the default; the warm seed is kept for cases where the cold start struggles to launch.

5.4 Metals and common finalization

In every family the metals start maximally neutral: the entire elemental abundance is placed in the neutral ground stage,

$$f_{X^0} = \frac{n_X/n_{\rm H}}{1+4Y}, \qquad f_{X^+} = f_{X^{++}} = 0 \tag{24}$$

(`set_IC.f90`, lines 152–162; abundances from `metals.inp`). The first ionization-equilibrium sweep redistributes each element over its stages immediately, so the choice costs at most one equilibrium solve. The complete IC (dimensional r, n, v, p, T) is written to `output/IC_dump.txt` before the first step.

5.5 Restarts (`Load IC`)

When `Load IC` is enabled, `set_IC` is bypassed and the previous run's profiles (including ghost cells) are read back from the `*_IC.txt` copies (`load_IC.f90`). This only sets the initial state; `Apply_BC` rewrites the ghost cells from the restored interior at the first step, so a restart cannot smuggle in a different boundary condition. (Operationally: the `output/` → `*_IC.txt` copy is performed by the GUI, not by the Fortran code: for scripted runs the `*_IC.txt` files must be seeded by hand, or the code stops on the empty files it creates.)

The restart *contract* that surrounds this read, the key `Restart intent:`, the eight-line metadata block both state files carry, its refusals, `Restart option change:` and the grid guard, is documented in `docs/composition_restart_and_base_handoff.p` Part 1, which is where a reader after that behavior should go. Two of its parts belong here because they are boundary statements. First, the base state a restart lands on is the reservoir this run resolved and not the one the file was written under: the `reservoir` field of the metadata block is compared (He/H and every trace element the run carries a reservoir for) and a disagreement outside `heh_dev_tol` refuses the load naming the element. Second, the grid belongs to the run and not to the state file: `verify_restart_radii` compares the cell centers of the physical cells $1..N$ to 10^{-10} relative and refuses a file written on any other grid, because `define_grid` builds the faces, the widths and the window indices $j_{\min}$, j_{flux} that the file does not carry, and a state cannot be marched on one grid's centers with another grid's faces. There is no interpolation onto a new grid.

A checkpoint restore inside a step is the one path that replaces the interior state without a flux update of its own, so `rebuild_state_from_checkpoint` rebuilds what is derived from the conserved state in one fixed order, composition first (because the pressure map and the boundary read it), then `Apply_BC` (so the stored base face state is the restored state's), then the primitive state and the temperature. Before that order was fixed, the stale base face state carried an O(H) error into one face of one Runge–Kutta stage, and the local error of the conservative rows now falls as H^4, the order of the three-stage scheme, instead of being dominated by that leak.

2026-08-13: a restart never carries zero metals into a metals-on base. The schema-2 writer emits the metal columns unconditionally, so testing only for the *presence* of a column accepted a file written by a metals-off run and let the restart proceed with zero metal density everywhere, while the lower boundary still counts the metals in the mass and particle budget under `eos_metals`, leaving an $O(1)$ residual in the first cells. `load_IC` now treats an element as absent in either case: no column for some stage, *or* columns that sum to zero over the interior. Such an element is rebuilt exactly as a cold start builds it, $n_X = a_X\, n_{\mathrm{H}}$ with all of it neutral, and the substitution is reported; the rebuild happens *before* `calc_rho` reconstructs the mass density, so the restart keeps the same mass closure the cold start has by construction. The elements that were substituted are recorded in `melem_from_abundance` and echoed by the setup report.

2026-08-15: a restart requires an IC written at the same N. Since the cell count is a runtime key (§1.), an IC file written at a different N would previously have died in the middle of the read with a bare end-of-file error. `load_IC` now counts the data records of `output/Hydro_ioniz_IC.txt` first (the files carry one record per cell, $N+2N_g$ rows), and stops with an explicit message naming both counts when they disagree, so a mismatched `Grid cells:` and a truncated file are reported rather than guessed at. There is no interpolation onto a new grid: a restart requires an IC written at the

same N.

2026-08-27: a restart lands on the reservoir the handoff states. A restart file carries the metal densities of the state it was written from. When the reservoir itself has moved since (which is what a flux-closure iteration does to the elemental ratios of the lower atmosphere), the base boundary condition would use the new El/H while the loaded column above it still held the old one, and the elemental budget $n_{\rm El}/n_{\rm H} = ({\rm El/H})_{\rm resolved}$ could not close above the first cell. `load_IC` therefore rescales the whole column of an element *the handoff itself states* by the single factor

$$r_{\rm El} \;=\; \frac{({\rm El/H})_{\rm handoff}}{({\rm El/H})_{\rm loaded,\ base\ cell}}, \tag{25}$$

the same factor for every ionization stage, so the loaded ionization split and the radial shape survive and only the normalization changes; the factor and both ratios are printed. "States" means the one door `set_element_abundance` guards, which the `<El>_H_base` keys of `base.inp` and the elemental ratios of a lower-atmosphere profile both go through, so an abundance that came from `metals.inp` alone, and every restart with no handoff at all, leaves the loop doing nothing and is bit-for-bit the restart the earlier code produced. El/H at the base cell is counted in nuclei, the element summed over its stages against the hydrogen nuclei (free plus the H bound in H_2, H_2^+, H_3^+ and HeH^+). Helium keeps its own convention (the base cells are set to the input He/H while the column above may carry a diffused split), and is not rescaled here.

6. Summary

Quantity	Lower boundary (base face $r_{\rm edg}(0)$)	Upper boundary ($r = r_{\rm max}$)
Where it acts	at the face; ghosts are averages of it	ghost cells
Pressure p	reservoir: $p_{\rm res}$ carried to the face	isothermal hydrostatic continuation
Entropy	reservoir isentrope through $(p_{\rm base}, T_{\rm base})$	same T as cell N (implied)
Velocity v	from (C−): $v_b = v_i + (p_b - p_i)/(\rho_i c_i)$	zero-gradient copy $v_g = v_N$
Density ρ	reservoir, handed to the interior isentrope on reversal	continuation, same factor as p
Ionization, metals	equilibrium solve in ghosts (no BC)	equilibrium solve in ghosts (no BC)
Radiation	in radiative shadow (large column)	unattenuated stellar spectrum
Type	characteristic (p, s) reservoir	free outflow
Code	`base_boundary.f90`	`Apply_BC.f90`, `free_outflow_ghost`

Table 1: Boundary conditions at a glance, with every option off (the defaults). Ghost cells: $N_g = 2$ per side; the base condition is imposed at the face and the ghost cell averages are the volume averages of the hydrostatic isentrope through it, so the base reaches the interior both through the HLLC flux at $r_{\rm edg}(0)$ and through the reconstruction stencil that reads the ghosts. The number of reservoir conditions is set by the face Mach number (§2.1); the table's rows are the subsonic-inflow branch, which is where every intended operating point lies.

The lower boundary is best read as a single physical statement: *the wind is rooted in a lower atmosphere whose pressure and entropy at the base level are known, and everything else at the boundary is what the interior sends down there.*

Its dynamical signature in a marching run is the breathing oscillation of §2.9, which is excluded from the convergence diagnostics through $r_{\rm esc}$ (and measured separately below that radius), differentiable for the Newton solver through the C^1 branch handover, and guarded against in the advection post-processor. The upper boundary is a free-outflow continuation whose accuracy matters most where the outer edge is subsonic.

The initial conditions are built to respect the same base state: all three families (§5.) match ($\rho_{\rm bc}$, $n_{\rm tot,bc}$+$\delta p_{\rm bc}$, v=0) at the base exactly, and the benchmark experience condenses into one rule: *the density column must be one the base can support*. The cold hydrostatic atmosphere (default) satisfies this trivially and evaporates into the wind on its own; the transonic IC replaces it with a flowing solution where the hydrostatic start cannot launch (deep-RLOF, sonic point at L1); the hot-Parker warm seed may overlay temperature, ionization, and velocity on the cold density, but seeding the Parker *density* itself tips the breathing base from self-limiting to self-amplifying and crashes the run.

7. Empirical test matrix of the retired ghost closure (HD209458b)

What this section is, and what it is not. The matrix below was run in 2026-08, before the characteristic face condition of §2. replaced the component-wise ghost closure. Its rows exercise the base velocity and ghost-pressure options that no longer exist (§2.8), so it is a record of how that closure behaved and *not* a description of the present boundary: none of its $\dot{M}$ values or convergence verdicts can be reproduced by today's binary, which refuses those keys outright. It is kept because two of its conclusions outlived the closure and are still the operating rules of this code, the flux-based convergence criterion of §7.1 and the assessment of the Wind-AE starter in §7.2, and because the base-anchor and Shapiro rows are the only measurements of those two surviving keys on this configuration.

7.1 Convergence criterion: history and the flux-based decision

The convergence test used here changed over the course of this work, and it is worth stating plainly where it landed and why.

- **Before the Newton finish: flux criterion.** The original time-marching run (PLM, then two-stage PLM→WENO3) was stopped on the *flux* criterion: the fractional variation of the mass flux, $\Delta\dot{M}/\dot{M} < 10^{-3}$, where $\dot{M} \propto \rho v r^2$ (the code's du is precisely the radial spread of $\rho v r^2$ over the wind). This is the ATES criterion of Caldiroli et al. (2021) and is equivalent to the CETIMB requirement (Koskinen et al. 2013a) that the flux constant $F_c = \rho v r^2$ be constant with altitude.

- **With the Newton finish: a stronger residual criterion.** When the JFNK steady solver was added, it was driven to a much stricter target: the steady residual $\|R\| = \|{\rm d}u/{\rm d}t\| < 10^{-3}$ on *every* conserved equation (mass, momentum, energy). This residual norm is far more demanding than flux flatness, and at the time these runs were made the JFNK solver reached it on only some of them; several full-physics runs were labeled "JFNK fail" even though their wind was perfectly flux-converged. *(2026-08-10: the "JFNK fail" outcomes recorded in this section were produced with a diagonal scaling whose momentum floor was set by the base cell, and with a watchdog that aborted after 15 iterations without a new best $\|R\|$, before the opening pseudo-transient has finished raising*

it. Both have since been replaced and the same configurations converge. The residual-vs-flux decision *below is unaffected, but the "held up by the near-base momentum" attribution attached to individual rows is withdrawn.)*

- **The reference codes use only the flux criterion.** Neither ATES (Caldiroli 2021: $\Delta\dot{M}/\dot{M} < 10^{-3}$) nor CETIMB (Koskinen 2013a: F_c constant with altitude) imposes a residual-norm threshold. The reference standard for "the wind has converged" is flux flatness; the residual norm is an EXHALE addition tied to the optional Newton finish.

- **Decision (adopted).** The **convergence is judged by the flux criterion**, matching the reference codes; the steady residual $\|R\|$ is computed and reported *for reference only* and gates the stop solely when the user explicitly requests it (`Resid tol:`). In the code the default monitor is flux-based ($du < du_{\rm th}$ plus mass-flux level stability) and the residual, volume-weighted by default (`resid_vol` in `parameters.f90`), since the L-∞ max is inflated by the small near-base cells, is printed each `N_resid` steps as a diagnostic. The `verdict` column of the table below therefore reads `flux-conv`, not "JFNK fail", for runs that are flux-converged but whose residual finish did not reach the target.

- **2026-08-11: the du triggers are armed on an ascending crossing.** du is the radial spread of $\rho v r^2$, so a *small* du is evidence of relaxation only for a state the marching loop has actually relaxed. A freshly generated IC (cold hydrostatic, transonic, Wind-AE, hot Parker) is an analytic profile whose du measures the smoothness of the formula rather than the flatness of a wind: the WASP-121 b transonic IC measured $du(1) = 1.85 \times 10^{-5}$, which tripped both the Newton hand-off and the secondary-ionization flip on step 1 and ran away to NaN. Each du-keyed trigger (the $du < du_{\rm th}$ stop, and the JFNK hand-off that the secondary-ionization flip shares) is therefore *armed* only once du has been seen at or above its own threshold, and fires only on the way back down (`du_stop_armed`, `du_newton_armed` in `EXHALE_main.f90`). A state read back with `Load IC? True` was relaxed by the marching loop that wrote it, so its du is meaningful and both triggers start armed. Cold hydrostatic starts arm on step 1 anyway (measured $du(1) = 0.37$ for the WASP regression cases), so their behavior is unchanged.

- **2026-08-15: the Wind-AE IC measured under that guard.** The arming guard had been measured on the transonic and cold-hydrostatic ICs but not on `IC mode: windae`, whose $\sim$3-step false du stop is what originally forced the "always Newton-finish" rule. Measured on `examples/12_windae_ic_hd209` with a 2000-step cap: du(step 2) $= 1.43 \times 10^{-4}$, below every threshold on the untouched generated IC, and nothing fires; the du stop arms at step 23 ($du = 1.027 \times 10^{-3}$) and the Newton hand-off at step 222 ($du = 1.002 \times 10^{-2}$), both on the first ascending crossing. The false stop is structurally gone for this IC mode too. This does *not* retire the rule of thumb: a du-stop $\dot{M}$ still carries the usual path-dependent spread, so the Newton finish stays the prescription for a quantitative rate.

2026-08-11: the tabulated rates are a product of the solver and the cooling as they stood when the matrix was run. Two changes since then move the numbers in the two tables below, and neither has been re-measured on this matrix. (i) The JFNK line search now accepts, stores, and judges convergence on the true residual (with the WENO smoothness weights recomputed at each trial state rather than frozen at the previous iterate), so runs that were recorded here as failing their residual finish are not evidence about the current solver. (ii) The two ground-term fine-structure

coolants of the time (the set has since grown to the eight C I/C II/N II/O I lines) are allowed to be optically thick, and the CHIANTI coronal fits are switched off smoothly below their 10^3 K validity floor; on a planet whose base falls out of the coronal regime this changes the base equilibrium temperature by hundreds of kelvin. The tables are kept as the dated record of the base-BC comparison they were made for, and the *relative* ordering across base modes is what they are being cited for. They should not be read as current mass-loss rates: the production HD 209458 b run under the present code gives $\log_{10}\dot{M} = 9.47$ at $\|R\| = 1.8 \times 10^{-4}$ (the 2026-08-19 production set; it read 9.31 at 7.7×10^{-4} when this paragraph was written on 2026-08-11). These runs were not repeated, because repeating them is a re-measurement of the whole matrix rather than a correction to it.

To pin down how the lower-boundary choices above actually affect a converged wind, HD209458b was relaxed under a controlled combination of solver and base options (the runs live in `EXHALE/HD209458b_test/`). Each folder name encodes the model as `<base>_<solver>_He{0,1}_met{0,CNO,FULL}[_flags]`: **base** is `dens` ($n_0 = 10^{14}\,\mathrm{cm}^{-3}$ fixed; the `dens12p5` run lowers it to $\log n_0 = 12.5$) or `pres1ub` (pressure-anchored at 1 μbar, §2.11); **solver** is `PLM` (single-stage, stopped at a *du* plateau), `PWN` (two-stage PLM→WENO3 then a Newton/JFNK finish), or `PLMN` (single-stage PLM then JFNK, skipping WENO3); **S/V** are the Shapiro filter (§2.11) and the retired mass-flux base velocity (§2.8); **He/met** are He 2^3S and the trace metals (C,N,O or the full C,N,O,Mg,Ca,Na,Fe set); **IC** is the cold hydrostatic or the Wind-AE warm start (§5.). The results are split into a pressure-anchored-base table (Table 2) and a density-anchored-base table (Table 3).

In both tables: $\dot{M}$ is in $\mathrm{g\,s^{-1}}$, v_{out} in $\mathrm{km\,s^{-1}}$, and neg.% is the *radius-weighted* infall fraction, the fraction of the linear radius range $[r_1, r_N]$ over which $v < 0$ (each cell weighted by its width Δr). This is used in place of the raw cell-count fraction, which the non-uniform grid (cells packed near the base) inflates: the dense near-base sliver of sub-$10\,\mathrm{m\,s^{-1}}$ noise would otherwise read as tens of percent. The radius-weighted measure cleanly separates that base noise ($\lesssim 4\%$) from genuine infall (tens of percent). "-" marks a run on an infall branch (wind $\dot{M}$ undefined).

The **verdict** column reports convergence *by the flux criterion* (the $\Delta\dot{M}\%$ column), *not* by the residual: `Newton` = a genuine Newton-converged steady state (solver `info=0`, $\|R\| < 10^{-3}$); `flux-conv` = flux-converged ($\Delta\dot{M}\% < 1$) but the JFNK residual finish did not reach $\|R\| < 10^{-3}$: by the ATES/CETIMB standard these *are* converged winds, not failures; `marginal` = $\Delta\dot{M}\% \sim$1–5% (a wind, not tightly flat; carries the ∼2–5% path-dependent $\dot{M}$ spread of the PLM-stall runs); `not conv` = $\Delta\dot{M}\% > 5\%$ (an unconverged / breathing wind); `infall` = relaxed to an accretion branch.

$\Delta\dot{M}\%$ is the radial spread of the mass flux $\rho v r^2$ over the wind region ($r \geq r_{\mathrm{esc}}$), (max−min)/mean. This is the actual **convergence criterion of the reference codes**: ATES (Caldiroli 2021) stops at $\Delta\dot{M}/\dot{M} < 10^{-3}$, and CETIMB (Koskinen 2013a) requires $\rho v r^2 = F_c$ to be constant with altitude. We mark $\lesssim 1\%$ as *flux-converged* (a genuine steady wind *by the reference standard*, independent of the $\|R\|$/Newton outcome); "-" = infall (not a wind). Note that all `PWN` rows are flux-converged ($\leq 0.9\%$) even where the stricter $\|R\|$ test labels them "JFNK fail". The "JFNK fail" labels date from the pre-2026-08-10 solver; the flux verdicts do not depend on them.

folder	BC	solver	S	V	He	met	IC	$\log \dot{M}$	$v_{\rm out}$	neg.%	$\Delta\dot{M}$%	verdict
`pres1ub_PWN_He0_met0_cold`	pres	PWN	–	–	–	–	cold	10.39	+8.59	0.0	0.53	Newton
`pres1ub_PWN_He1_metCNO_cold`	pres	PWN	–	–	✓	CNO	cold	9.53	+6.63	1.6	0.62	flux-conv
`pres1ub_PWN_He1_metFULL_cold`	pres	PWN	–	–	✓	FULL	cold	9.37	+6.29	2.2	0.87	flux-conv
`pres1ub_PWN_He1_metFULL_cold_duJFNK`	pres	PWN$^{(d)}$	–	–	✓	FULL	cold	9.37	+6.29	1.7	0.88	flux-conv$^{(d)}$
`pres1ub_PLMN_He1_metFULL_cold`	pres	PLMN	–	–	✓	FULL	cold	9.40	+6.10	1.8	3.5	marginal§
`pres1ub_PWN_He1_metCNO_cold_Vflux`	pres	PWN	–	✓	✓	CNO	cold	9.53	+6.63	1.6	0.74	flux-conv
`pres1ub_PWN_He1_metCNO_cold_Shap`	pres	PWN	✓	–	✓	CNO	cold	10.21	+7.45	25.3	114	no conv.†
`pres1ub_PLM_He0_met0_cold`	pres	PLM	–	–	–	–	cold	10.40	+8.36	0.0	3.5	marginal

Table 2: **Pressure-anchored base** ($1\,\mu$bar, $n_0 \approx 5\times10^{12}\,\mathrm{cm}^{-3}$). Columns and general notation as described in the text. †The pressure + Shapiro run never converged: it ran to the 10^6-step cap with 25% of the radius in infall (vs. 1.6% for the same run without Shapiro), so the filter both prevents convergence and deepens the infall; the quoted $\dot{M}$ is the outer-region value, not a steady state. §`PLMN` = single-stage PLM → JFNK (the WENO3 stage skipped). It reaches an *essentially identical final state* to the `PWN` (PLM→WENO3→JFNK) row above ($\log\dot{M}$ = 9.40 vs. 9.37, $v_{\rm out}$ = +6.10 vs. +6.29, overlapping $T/v/n/p$ profiles) in about *half* the steps (9.4×10^4 vs. 1.7×10^5); both still fail JFNK and stall, so the WENO3 stage is not the JFNK bottleneck. $^{(d)}$The `_duJFNK` row is the same full-metal case but with the *flux*-keyed JFNK hand-off (`Solver: Newton`, JFNK fired once the wind reached $du < 10^{-2}$ rather than $\|R\| < 0.05$). With this hand-off *plus* the volume-weighted residual norm the JFNK drove $\|R\|$ down to 2.3×10^{-3} (a $\sim$100× improvement over the 0.17 it reached with the $\|R\|$-keyed hand-off), but it stalled there ($\|R\|$ would not drop below $\approx 2\times10^{-3}$ for 15 iterations, the holdout being the base cell j=1 momentum), so it still falls back and the *final wind is identical* to the `metFULL` PWN row above ($\dot{M}$, $v_{\rm out}$, $\Delta\dot{M}$% all unchanged). It confirms that the only barrier to a genuine $du < 10^{-3}$ / $\|R\| < 10^{-3}$ convergence for full physics is the localized near-base momentum (§2.9), not the solver path.

folder	BC	solver	S	V	He	met	IC	$\log \dot{M}$	$v_{\rm out}$	neg.%	$\Delta\dot{M}$%	verdict
dens_PWN_He0_met0_cold	dens	PWN	–	–	–	–	cold	10.50	+8.80	0.0	0.41	Newton
dens12p5_PWN_He0_met0_cold	dens	PWN	–	–	–	–	cold	10.38	+8.58	0.0	0.42	Newton
dens_PWN_He1_metCNO_cold	dens	PWN	–	–	✓	CNO	cold	9.61	+6.79	2.3	0.54	flux-conv
dens_PWN_He0_met0_windae_spherical	dens	PWN	–	–	–	–	windae	10.37	+5.77	1.4	55	not conv‡
dens_PLM_He0_metCNO_cold	dens	PLM	–	–	–	CNO	cold	9.72	+6.77	3.9	4.2	marginal
dens_PLM_He0_met0_cold_Vflux	dens	PLM	–	✓	–	–	cold	10.50	+8.58	0.0	3.5	marginal
dens_PLM_He0_met0_cold_Shap	dens	PLM	✓	–	–	–	cold	-	−13.34	100	-	infall
dens_PLM_He0_met0_cold_Shap_Vflux	dens	PLM	✓	✓	–	–	cold	-	−13.32	100	-	infall
dens_PLM_He0_met0_windae_spherical	dens	PLM	–	–	–	–	windae	10.08	+3.68	0.6	5.1	not conv
dens_PLM_He0_met0_windae_spherical_Shap_Vflux_Force	dens	PLM	✓	✓	–	–	windae+F	-	−20.03	95	-	infall

Table 3: **Density-anchored base** ($n_0 = 10^{14}\,\mathrm{cm}^{-3}$; dens12p5: $\log n_0 = 12.5$). Columns and general notation as described in the text. ‡Wind-AE warm start: JFNK never triggered ($\|R\|$ stayed above the switch); the Wind-AE IC marched under WENO3 drifts to an unconverged breathing state ($du \sim 0.7$), mostly outflow: it does *not* give a clean Newton solution.

What the matrix shows.

- **The Shapiro filter drives infall.** On the cold-IC density runs, every S-on case relaxes to a global infall ($v < 0$ over $\approx 100\%$ of the radius) while the matched S-off case is a clean outflow, independent of the velocity BC: the headline reason the filter is opt-in (§2.11). On the pressure + full-physics case the filter instead prevents convergence altogether (row †).

- **The base BC barely changes the converged wind: the lever is the base *density*.** With a Newton finish, density ($n_0 = 10^{14}$) and pressure ($1\,\mu$bar) bases give nearly the same $\dot{M}$ (10.50 vs. 10.39). A *density*-anchored run at the same low density (`dens12p5`, $\log n_0 = 12.5$) reproduces the pressure result almost exactly (10.38 vs. 10.39; +8.58 vs. +8.59), confirming that pressure-anchoring acts only through the lower base density, not through the anchoring variable itself. The mass-flux velocity (V) was benign under that closure (it reproduced the valve result); both are retired keys today (§2.8).

- **Physics lowers $\dot{M}$.** He 2^3S + C/N/O cooling drops $\log\dot{M}$ from $\sim$10.4 to $\sim$9.5–9.6; the full metal set (adding Mg/Ca/Na/Fe line cooling) lowers it further to 9.37. With full physics the JFNK solve fails for both bases and falls back to a stall.

- **Skipping WENO3 before the Newton finish (`PLMN`) changes nothing but the cost.** On the full-metal case, `PLMN` (PLM→JFNK) reaches an essentially identical final state to `PWN` (PLM→WENO3→JFNK) ($\log\dot{M} = 9.40$ vs. 9.37, overlapping profiles) in about half the steps, and both still fail JFNK and stall. So the reconstruction path is not the JFNK bottleneck; the stiff coupled (He 2^3S + 7-metal) residual is. A genuine Newton finish for full physics would need a smoother residual (tighter chemistry tolerance) or a relaxed target, not a different reconstruction.

- **The Wind-AE warm start does not help here.** It either du-stops at step 2 (returning $\approx$ the IC) or, with a Newton finish, never triggers JFNK and drifts to an unconverged breathing state. The cold hydrostatic IC, consistent with EXHALE's own base, is the reliable choice.

7.2 Assessment: does the Wind-AE starter help?

So far, for HD209458b, the Wind-AE starter does not help: the cold IC is better. Across every Wind-AE-seeded run in Table 3 the outcome is one of: a *du*-stop at step 2 that simply returns the seed (no real relaxation), an unconverged breathing state when a Newton finish is requested (JFNK never triggers), or, when marching is forced, a drift to infall ($v_{\rm out} = -20\,{\rm km\,s^{-1}}$). An earlier scratch run converged instead to a global *infall root*. None reached a clean, Newton-converged outflow, whereas the cold IC with a `PWN` finish does (solver `info=0`, clean wind).

Why. A *starter* replaces the entire state, *including its own base*, with another code's solution (§5.). That imported base is generally inconsistent with EXHALE's fixed lower boundary ($\rho_{\rm bc}$, T_0); the mismatch perturbs the near-base flow off the transonic-wind saddle, and since that saddle is unstable, the run is pulled to the nearby attractors: infall, or the breathing limit cycle. This is the same dense-base/lower-BC tension that drives the breathing (§2.9), now triggered by the seed rather than by a bad global IC.

In fairness. This does *not* make Wind-AE useless. It was brought in for the *hard* cases where a cold start fundamentally cannot launch a wind: deep Roche-lobe-overflow geometries (e.g. WASP-121b Case D, with the sonic point essentially at L1) and low-gravity boil-off planets, where it is, at present, the only way to obtain a transonic seed. HD209458b is an easy case for which the cold IC already works, so there the starter is both unnecessary *and* mildly harmful through the base mismatch. The honest statement is therefore: *the starter is unnecessary for easy cases and, for its intended hard cases, has not yet delivered a clean solution because the base-BC reconciliation is unsolved*, not that the approach is without value.

What would make it useful. The base mismatch must be removed. Two plausible routes (untested): (i) have the Wind-AE bridge hand off only a *warm overlay* (temperature, ionization, and velocity) on top of EXHALE's own hydrostatic density and base, i.e. demote the "starter" to a "warm seed" (§5.) that stays consistent with the lower boundary; or (ii) ramp the imported base onto EXHALE's $(\rho_{\rm bc}, T_0)$ boundary before releasing the hydrodynamics. Either could keep the transonic-seed advantage Wind-AE offers for the RLOF cases while avoiding the infall/breathing seen here.

References

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